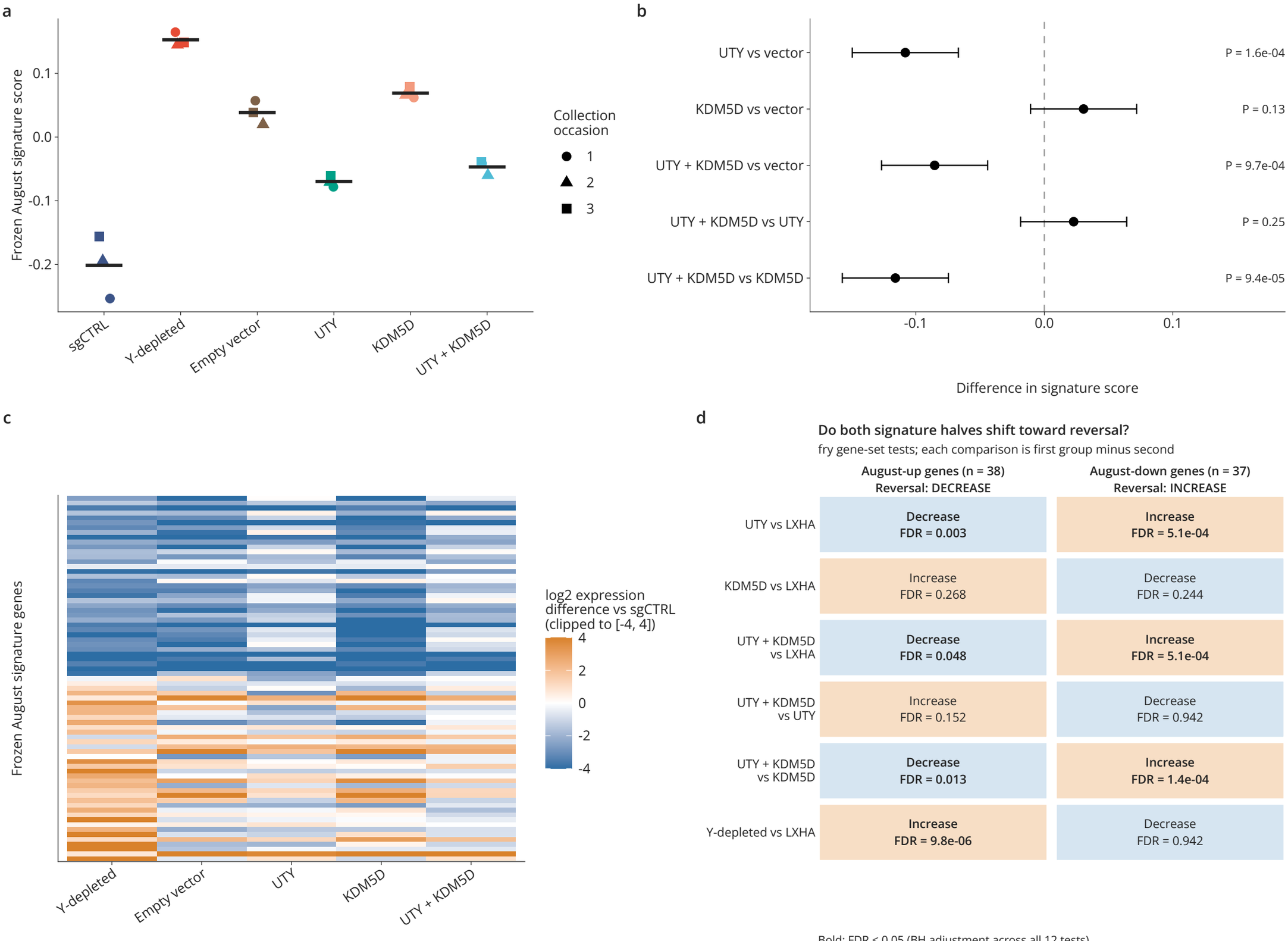

a
Frozen August signature score
0.1
0.0
-0.1
-0.2
sgCTRL
Y-depleted
Empty vector
UTY
KDM5D
UTY + KDM5D
Collection occasion
1
2
3
b
UTY vs vector
KDM5D vs vector
UTY + KDM5D vs vector
UTY + KDM5D vs UTY
UTY + KDM5D vs KDM5D
P = 1.6e-04
P = 0.13
P = 9.7e-04
P = 0.25
P = 9.4e-05
-0.1
0.0
0.1
Difference in signature score
c
Frozen August signature genes
Y-depleted
Empty vector
UTY
KDM5D
UTY + KDM5D
log2 expression difference vs sgCTRL (clipped to [-4, 4])
4
2
0
-2
-4
d
Do both signature halves shift toward reversal?
fry gene-set tests; each comparison is first group minus second
August-up genes (n = 38) Reversal: DECREASE
August-down genes (n = 37) Reversal: INCREASE
UTY vs LXHA
Decrease FDR = 0.003
Increase FDR = 5.1e-04
KDM5D vs LXHA
Increase FDR = 0.268
Decrease FDR = 0.244
UTY + KDM5D vs LXHA
Decrease FDR = 0.048
Increase FDR = 5.1e-04
UTY + KDM5D vs UTY
Increase FDR = 0.152
Decrease FDR = 0.942
UTY + KDM5D vs KDM5D
Decrease FDR = 0.013
Increase FDR = 1.4e-04
Y-depleted vs LXHA
Increase FDR = 9.8e-06
Decrease FDR = 0.942
Bold: FDR < 0.05 (BH adjustment across all 12 tests).
Direction and significance are distinct; color indicates direction only.